NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Marks: 20	**QUIZ-4**	**Time: 20 Minutes**

Subject: Probability and Statistics **Section: D**
Instructor: Hazber Samson Date: 01-04-2026
Student Name_____ Roll No_____

Q1. **(3+3+4=10)**

The random variable X is $B(10, p)$ where $p < 0.5$ The variance of X is 1.875. Find
(a) the value of p
(b) $E(X)$
(c) $P(3 \leq X \leq 5)$

Q2. **(3+3+4=10)**

The average number of calories in a 1.5-ounce chocolate bar is 225. Suppose that the distribution of calories is approximately normal with s 10. Find the probability that a randomly selected chocolate bar will have

a. Less than 200 calories
b. more than 215 calories
c. Between 200 and 220 calories